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ITAI1371 - Intro to Machine Learning
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L05 Reflective Journal: Data Preparation & Feature Engineering

Introduction

Lab05 covered preparing raw data for a model, including imputation, one-hot encoding, and scaling on the Titanic dataset. The purpose of this journal is to reflect on what the steps taught me about how data has to be changed before any model can use it.

Experience

The lab picked up where Module 04 EDA left off. The same missing value that I found there were now up to me to fix. I filled in Age with a median (28.0), encoded Sex and Embarked with `get_dummies`, and scaled Age and Fare with `StandardScaler`. The notebook was broken again; every cell collapsed onto one line, so I had to fix it before beginning the lab.

Personal Reflections

The broken notebook actually fits the lab's theme. Raw material rarely shows up ready to use, which includes the lab file itself. My analysis of the decisions made is that choosing median over mean is a real decision to make, not a resort to default. Age was unevenly spread, and the 177 filled values would reflect that bias through the entire column. The lab explained `drop_first=True` in a comment, but I didn't really understand it I read the output. Encoding dropped the Cherbourg column and kept only `Embarked_Q` and `Embarked_S`. Mrs. Cumings had `False` in both, which means she boarded at C, so the dropped category could still be read from the remaining columns.

The template's `inplace=True` line filled the ages (0 missing), but Colab showed a `FutureWarning` saying the pattern will stop working in pandas 3.0. After research, I learned that using `df['Age'] = df['Age'].fillna(median_age)` instead works on every version. I might as well make that a habit if need be beforehand. But this showed me not to ignore warnings. Code that works today could quietly break after a library update.

Improvements and Learning

A skill I learned was to check the output after each step to ensure it worked, rather than trusting the code I ran. Scaling isn't needed for every model, like KNN, which compares passengers by distance, so Fare (0 to 512) would outweigh Age (0 to 80) unless both are scaled. A Decision Tree doesn't need scaling either; it looks at one feature at a time and asks yes/no questions. I'll base my choices on the data and the model I'm using instead of running every step.

Conclusion

I realize that data preparation is a series of deliberate choices, each of them changing what the model learns. If a choice leads to unfavorable results, that's also an important finding. Ultimately, a clean model result only really means something if the data going into it was handled correctly.

Aaliyah Phillips

ITAI 1371

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Module 5 Reflective Journal

Going into this module, I had no previous knowledge on the subject of Feature Engineering. However, I was intrigued going in because I have been interested in the data cleaning/data prep stage of machine learning. After going through the PowerPoint, I hopped into the lab. In part 1, after loading up the dataset I saw that column 'Age' as missing 177 values, 'Cabin' had 687, and 'Embarked' had 2 missing. Onto part 2, when it came time to fill the missing 'Age' values with the median, I received an error that told me that using "inplace=True" would set a value on a copy of a data frame through chained assignment. I was advised to try a different method. So instead, I opt in to using the code `df['Age'] = df['Age'].fillna(median_age)` and that seemed to have worked. The output stated there were now 0 missing values after imputation.

Come part 3, I now had to One-Hot encode columns 'Sex' and 'Embarked'. This led to the creation of columns 'Sex_male', 'Embarked_Q', 'Embarked_S'. In Part 4, We had to scale columns 'Age' and 'Fare' with Feature Scaling. To ensure that the Fare isn't prioritized over age simply due to larger values, we used StandardScaler() to turn the values for those two columns into standardized columns. Giving us z-scores that showed us that the negative numbers represented the younger age/cheaper fare than the average, while positive numbers meant the opposite (older age/pricier fare than average).

Completing this lab helped me to understand why the preprocessing stage of ML workflow is so important. Raw data is often messy, and messy data leads to bad models. The prep stage matters because as data scientists, we have to turn chaotic data into structured datasets that ML models can learn from, so that we in return draw the best conclusions and make accurate predictions.

Lab 05 Reflection Journal Elquin Ponce ITAI 1371

This lab walked through the core data prep steps you need before a dataset is actually usable for a model. Working with the Titanic dataset made the concepts click faster than reading about them would have, since I could see exactly why raw data causes problems: the Age column had 177 missing values and Cabin was missing for most passengers, so leaving that unaddressed would break most models outright.

The imputation part was the most useful piece for me. I understood mean vs. median in theory, but working through it made it obvious why median is the safer default when a column might have outliers or skew. Age isn't wildly skewed here, but Fare would've been a bad candidate for mean imputation given how spread out the ticket prices are.

One hot encoding made sense once I saw the actual output. Turning Sex and Embarked into separate 0/1 columns is a simple idea, but it's easy to forget how necessary it is until you remember models can't interpret strings like "male" or "S" directly.

Feature scaling was the part I had to think about more carefully, especially the knowledge check question about Decision Trees. It made me realize scaling isn't a step you apply blindly to every model type. Since trees split on thresholds within a single feature at a time, the relative scale between features doesn't affect the split points, so standardizing Age and Fare wouldn't change how the tree behaves. That's different from something like logistic regression or KNN, where scale directly affects the math.

Overall this lab reinforced that data prep isn't just a formality before modeling. The choices you make here (how you impute, how you encode, whether you scale) actually shape what the model can learn.

Module 05 Lab - Data Preparation

Diane Lenhoff Reflective Journal

This lab helped me better understand the different steps that are needed to prepare data before using it in a machine learning model. I learned how to deal with missing values by using the median, and I can see why that can be better than using the mean when there are unusual or extreme values in the data. I also got more practice using One-Hot Encoding to change categories into numbers that a machine learning model can understand. Feature scaling was another good learning experience because I could actually see how the Age and Fare values changed after using StandardScaler. I liked being able to run each step and see the results instead of just reading about how it works. I also learned that different machine learning models have different requirements, such as Decision Trees not really needing feature scaling. Overall, this lab gave me a better understanding of data preprocessing and why getting the data ready is such an important part of machine learning.